GSM-NoOp

Oliver picks 44 kiwis on Friday. Then he picks 58 kiwis on Saturday. On Sunday, he picks double the number of kiwis he did on Friday, but five of them were a bit smaller than average. How many kiwis does Oliver have?

---

o1-mini: Oliver picks kiwis over three days as follows:
- Friday: 44 kiwis
- Saturday: 58 kiwis
- Sunday: Double the number he picked on Friday, which is $2 \times 44 = 88$ kiwis

However, on Sunday, 5 of these kiwis were smaller than average. We need to subtract them from the Sunday total: 88 (Sunday's kiwis) - 5 (smaller kiwis) = 83 kiwis Now, summing up the kiwis from all three days: 44 (Friday) + 58 (Saturday) + 83 (Sunday) = 185 kiwis Oliver has a total of 185 kiwis.

Llama3-8B: Let's break it down step by step:
Oliver picks 44 kiwis on Friday. He picks 58 kiwis on Saturday. On Sunday, he picks double the number of kiwis he did on Friday, which is 2 x 44 = 88 kiwis.
Five of the kiwis picked on Sunday are a bit smaller than average, so we subtract 5 from the total number of kiwis picked on Sunday: 88 - 5 = 83 kiwis. Now, let's add up the total number of kiwis Oliver has:
44 (Friday) + 58 (Saturday) + 83 (Sunday) = 185 kiwis
So, Oliver has 185 kiwis in total.

Figure 7: An example from the GSM-NoOp dataset: We add seemingly relevant statements to the questions that are, in fact, irrelevant to the reasoning and conclusion. However, the majority of models fail to ignore these statements and blindly convert them into operations, leading to mistakes.

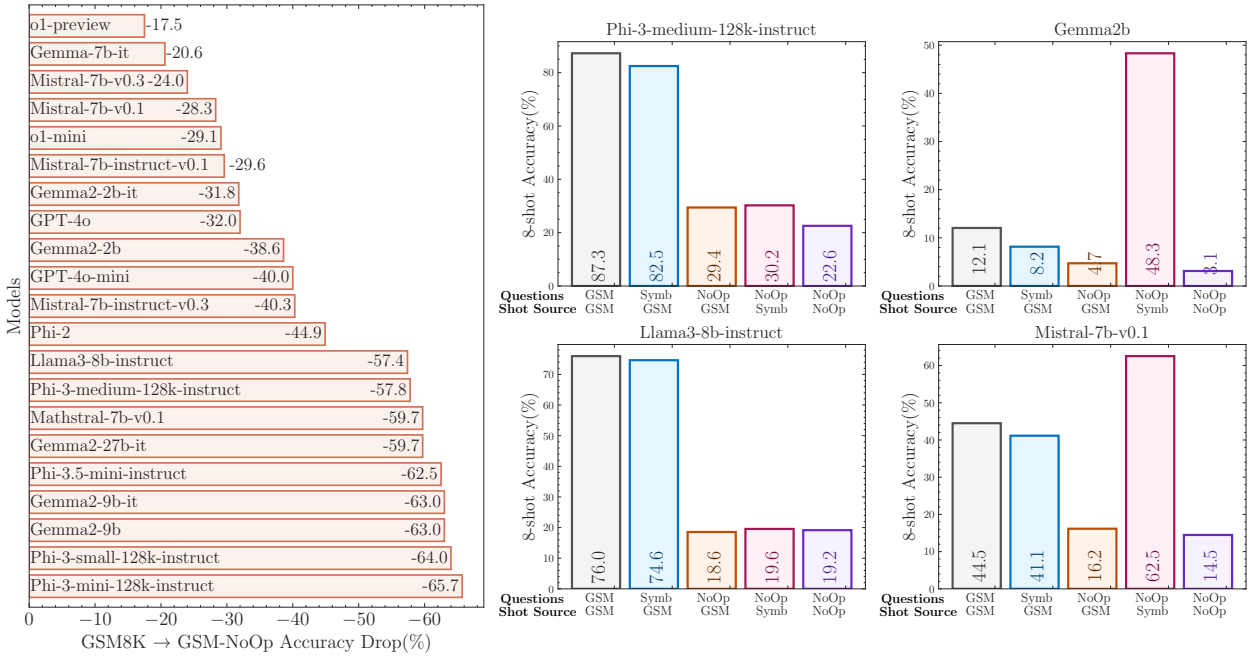
4.4 Can LLMs Really Understand Mathematical Concepts?

In the previous sections, we studied the impact of *type of change* and *difficulty* on the performance distribution. In this section, we demonstrate that models are susceptible to catastrophic performance drops on instances not part of the training distribution, potentially due to their reliance on in-distribution pattern-matching.

We introduce GSM-NoOp, a dataset designed to challenge the reasoning capabilities of language models. To create the templates, we add seemingly relevant but ultimately inconsequential statements to GSM-Symbolic templates. Since these statements carry no operational significance, we refer to them as *"No-Op"*. These additions do not affect the reasoning required to solve the problem.

Fig. 7 illustrates an example from GSM-NoOp. An interesting observation is that models tend to blindly subtract the number of smaller fruits, potentially because their training datasets included similar examples that required conversion to subtraction operations. In the Appendix, we include additional failure cases from GSM-NoOp. Overall, we find that models tend to convert statements to operations without truly understanding their meaning. For instance, a common case we observe is that models interpret statements about "discount" as "multiplication", regardless of the context. This raises the question of whether these models have truly understood the mathematical concepts well enough. Consequently, as shown in Fig. 8a, there is a catastrophic performance decline across all tested models, with the Phi-3-mini model experiencing over a 65% drop, and even stronger models such as o1-preview showing significant declines.

To better understand this performance drop, we conducted another experiment. While our previous evaluations on GSM-P2 used the original 8-shots of GSM8K, here we explore two new scenarios where we change the source of the 8-shots. We report the results in Figures 8b and 8c.



(a) (b) (c)

Figure 8: **(a)** The performance of models drops significantly on `GSM-NoOp`, with more recent models experiencing a greater decline than older ones. **(b)** As previously demonstrated, performance on `GSM-Symbolic` is very close to that on `GSM8K`. However, on `GSM-NoOp`, the significant drop in performance cannot be recovered, even when using the exact same question's variation as shots (`NoOp-Symb`) or when using different questions with different `GSM-NoOp`that contain No-Op operations (`NoOp-NoOp`) as shots. **(c)** Notably, some models that perform significantly worse than those in (b) on `GSM8K` and `GSM-Symbolic` show much better performance on `NoOp-Symb`.

- **NoOp-Symb** (Using `GSM-Symbolic` shots of the *same* question): During evaluation, we include 8 different shots of the *same* question coming from `GSM-Symbolic`. Hence, each shot provides the required reasoning steps. The target question from `GSM-NoOp` then presents yet another variation of the same question that is different only in values and the added clause that is inconsequential. This setup should simplify the task by making it clear that the extra information in the target question is irrelevant. However, as shown in Fig. 8b, the performance remains within the standard deviation, even with 8 shots of the same question providing the reasoning chain. Interestingly, Fig. 8c shows that some models can perform significantly better, even though they don't perform nearly as well on `GSM8K` and `GSM-Symbolic`. We believe this is a very notable observation.
- **NoOp-NoOp** (Using `GSM-NoOp` shots of *different* questions): Here, we provide 8 shots chosen randomly from different questions of `GSM-NoOp` in the context. These questions share the common fact that the correct answer should ignore the No-Op statement. We observe that for the Llama-3-8B model, the performance remains the same compared to the original No-Op model, while for the Phi-3 model, performance slightly decreases.

5 Conclusion

In this work, we have investigated the reasoning capabilities of large language models (LLMs) and the limitations of current evaluations on GSM8K. We introduced GSM-Symbolic, a novel benchmark with multiple variants designed to provide deeper insights into the mathematical reasoning abilities of LLMs. Our extensive study reveals significant performance variability across different instantiations of the same question, challenging the reliability of current GSM8K results that rely on single-point accuracy metrics. We found that while LLMs exhibit some robustness to changes in proper names, they are more sensitive to variations in numerical values. We have also observed the performance of LLMs deteriorating as question complexity increases.

The introduction of GSM-NoOp exposes a critical flaw in LLMs' ability to genuinely understand mathematical concepts and discern relevant information for problem-solving. Adding seemingly relevant but ultimately inconsequential information to the logical reasoning of the problem led to substantial performance drops of up to 65% across all state-of-the-art models. Importantly, we demonstrate that LLMs struggle even when provided with multiple examples of the same question or examples containing similar irrelevant information. This suggests deeper issues in their reasoning processes that cannot be easily mitigated through few-shot learning or fine-tuning.

Ultimately, our work underscores significant limitations in the ability of LLMs to perform genuine mathematical reasoning. The high variance in LLM performance on different versions of the same question, their substantial drop in performance with a minor increase in difficulty, and their sensitivity to inconsequential information indicate that their reasoning is fragile. It may resemble sophisticated pattern matching more than true logical reasoning. We remind the reader that both GSM8K and GSM-Symbolic include relatively simple grade-school math questions, requiring only basic arithmetic operations at each step. Hence, the current limitations of these models are likely to be more pronounced in more challenging mathematical benchmarks.

We believe further research is essential to develop AI models capable of formal reasoning, moving beyond pattern recognition to achieve more robust and generalizable problem-solving skills. This remains a critical challenge for the field as we strive to create systems with human-like cognitive abilities or general intelligence.

Acknowledgments

The authors would like to thank Max Horton, Fartash Faghri, Moin Nabi, and Devi Krishna for the valuable feedback and support.

References

Marah I Abdin, Sam Ade Jacobs, Ammar Ahmad Awan, Jyoti Aneja, Ahmed Awadallah, Hany Awadalla, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Harkirat S. Behl, Alon Benhaim, Misha Bilenko, Johan Bjorck, Sébastien Bubeck, Martin Cai, Caio César Teodoro Mendes, Weizhu Chen, Vishrav Chaudhary, Parul Chopra, Allie Del Giorno, Gustavo de Rosa, Matthew Dixon, Ronen Eldan, Dan Iter, Amit Garg, Abhishek Goswami, Suriya Gunasekar, Emman Haider, Junheng Hao, Russell J. Hewett, Jamie Huynh, Mojan Javaheripi, Xin Jin, Piero Kauffmann, Nikos Karampatziakis, Dongwoo Kim, Mahoud Khademi, Lev Kurilenko, James R. Lee, Yin Tat Lee, Yuanzhi Li, Chen Liang, Weishung Liu, Eric Lin, Zeqi Lin, Piyush Madan, Arindam Mitra, Hardik Modi, Anh Nguyen, Brandon Norick, Barun Patra, Daniel Perez-Becker, Thomas Portet, Reid Pryzant, Heyang Qin, Marko Radmilac, Corby Rosset, Sambudha Roy, Olatunji